Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Cloud Data Platforms, Lakehouse Architecture, Agentic AI, Autonomous Workflows, Prompt Orchestration, MLOps, Machine Learning, Anomaly Detection, Time-Series Forecasting, Data Governance, Lineage, CI/CD, Data SRE, Dimensional Modeling.

Programming Languages & Tools: AWS (S3, Redshift, Glue, EMR, Athena), Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Apache Spark, PySpark, Python, SQL, Power BI, Tableau, Collibra, KIRO IDE, Copilot, XGBoost, Scikit-Learn, SARIMA.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Cloud Data & AI Platform Engineer — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** using KIRO IDE, prompt orchestration, and Copilot to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50%.
- **Engineered autonomous machine learning anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision across cloud data platforms, safeguarding downstream BI and pipeline integrity.
- **Built and deployed high-throughput distributed pipelines and dimensional models** utilizing AWS (S3, Redshift), Azure (ADF, Databricks, Synapse), Snowflake, and PySpark to crunch hundreds of terabytes with 99.9% data availability SLAs.
- **Implemented automated source-to-target reconciliation firewalls and data governance** integrating Collibra lineage and automated validation to slash operational turnaround by over 80% and ensure audit-ready compliance.
- **Engineered automated testing and evaluation frameworks for conversational AI platforms** such as SpectrumIQ, cutting manual QA turnaround by 50% and accelerating production release cycles in enterprise environments.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Built and deployed predictive machine learning models** using XGBoost, Random Forests, and Scikit-Learn to forecast customer behavior, delivering actionable intelligence that improved service reliability by 20%.
- **Architected advanced time-series SARIMA forecasting models** to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing operational response times by 20%.
- **Developed scalable feature engineering and inference pipelines** bridging raw transactional telemetry with machine learning models to support real-time decision-making and business intelligence dashboards.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Created executive Key Operating Metric (KOM) dashboards and semantic models** in Power BI (DAX) and Tableau, translating complex transactional telemetry into strategic corporate insights.
- **Designed scalable star and snowflake schema dimensional models** optimized for high-performance querying and seamless integration with downstream analytics and AI consumers.
- **Partnered with cross-functional stakeholders** to define data requirements, establish reliable KPI metrics, and accelerate enterprise-wide data-driven decision making.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.

- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.